

8 Environmental conditions such as light intensity affect plant physiology.

- (a) Use the letter **X** to identify a point on the sketch graph in Fig. 8.1 where light intensity is acting as a limiting factor on the rate of photosynthesis. [1]

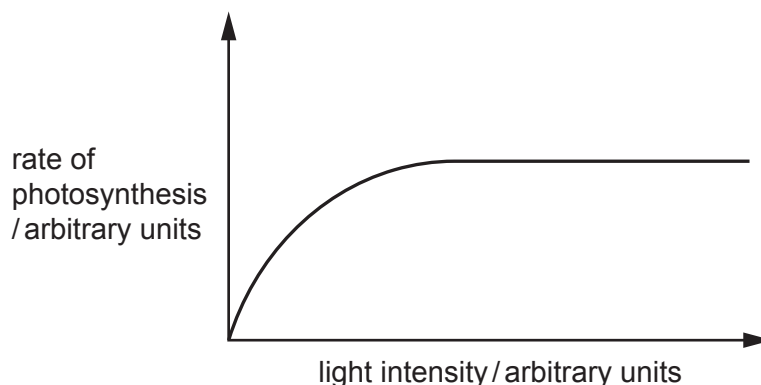


Fig. 8.1

- (b) An experiment investigated how light intensity affected gene expression in kale, *Brassica oleracea sabellica*.
- Two groups of kale plants were grown, with one group in high light intensity and one group in low light intensity. All other conditions were standardised.
 - After the same period of time, messenger RNA (mRNA) was extracted from each group of kale plants.
 - The mRNA was used to produce cDNA.
 - The cDNA was hybridised with probes for 89 621 kale genes on a microarray.
 - The microarrays showed which genes were switched on (expressed) in each set of conditions.

The results showed that expression of 18% of the genes was affected by light intensity.

- 14% of the genes were switched on only in high light intensity.
- 4% were switched on only in low light intensity.

- (i) State the name of the enzyme that produces cDNA from an mRNA template.

..... [1]

- (ii) State the name of the type of proteins that control gene expression in plants.

..... [1]

- (iii) Suggest why **more** genes were switched on only in plants growing in **high** light intensity compared to fewer genes that were switched on only in plants growing in low light intensity.

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..... [3]

[Total: 6]